1 Introduction

We draw on the human respiratory tract deposition model proposed by the International Commission on Radiological Protection (ICRP) to evaluate the deposition behavior of engineered bacteria in the human respiratory tract after inhalation. This model is based on extensive experimental data and theoretical derivations, and divides the respiratory tract into four main regions:

ET region (Extrathoracic): head airways (nose, pharynx, larynx)

BB region (Bronchial): trachea and bronchi (conducting zone)

bb region (Bronchiolar): bronchioles (transitional zone)

AI region (Alveolar-Interstitial): alveolar-interstitial region

The deposition mechanisms considered in the model include: **inertial impaction**, **gravitational sedimentation** and **diffusion**. These three mechanisms dominate the deposition behavior of particles in different size ranges.

2 Detailed derivation of deposition mechanisms and mathematical descriptions

Mathematical derivation of inertial impaction

Definition and physical meaning of the Stokes number

The core dimensionless parameter of the inertial impaction mechanism is the **Stokes number (Stk)**. It characterizes the strength of particle inertia relative to the characteristic fluid motion, defined as the ratio of the particle relaxation time (τ_p) to the characteristic flow time (τ_f):

$$Stk = \frac{\tau_p}{\tau_f}.$$

Particle relaxation time τ_p: the characteristic time required for a particle to adjust from an initial velocity to the fluid velocity. For spherical particles at low Reynolds numbers, the relaxation time is given by:

$$\tau_p = \frac{\rho_p d_p^2 C_c}{18\mu},$$

where ρ_p is the particle density, d_p is the particle diameter, μ is the dynamic viscosity of air, and C_c is a correction factor (approximately 1).

Characteristic flow time τ_f: the time required for the airflow to pass a characteristic length. In the respiratory tract, it is usually taken as the ratio of the airway diameter D to the characteristic flow velocity U:

$$\tau_f = \frac{D}{U}.$$

Therefore, the Stokes number can be written as:

$$Stk = \frac{\rho_p d_p^2 U C_c}{18\mu D}.$$

Physical meaning: When $Stk \ll 1$, the particle relaxation time is much smaller than the characteristic flow time; the particle can respond quickly to flow changes and almost completely follow the streamlines, making inertial impaction unlikely. When $Stk \gg 1$, the particle relaxation time is long and inertia is significant; the particle cannot follow the flow direction in time, thus deviating from streamlines and impacting the wall.

In the respiratory tract, the Stokes number is the dominant parameter describing inertial impaction. In the upper respiratory tract (e.g., nasal cavity, larynx, bronchial bifurcations), where the flow direction changes abruptly, U is relatively large and D is relatively small, so Stk is larger and inertial impaction mainly occurs here. Typically, when $Stk > 0.1$, inertial impaction begins to become important; when $Stk > 0.5$, the impaction efficiency increases significantly.

Particle motion equation

In the Lagrangian framework, the equation of motion for a single particle (neglecting gravity, Brownian forces, and considering only Stokes drag) is:

$$m_p \frac{d\vec{v}}{dt} = \frac{1}{B}(\vec{u} - \vec{v}),$$

where $B = \frac{1}{3\pi\mu d_p} C_c^{-1}$ is the particle mobility.

Inertial impaction efficiency

For airway bifurcations, a general form is:

$$\eta_{\text{imp}} = 1 - \frac{2}{\pi}\left[\sqrt{1-\left(\frac{\theta}{Stk}\right)^2} - \frac{\theta}{Stk}\arccos\frac{\theta}{Stk} + \arcsin\frac{\theta}{Stk}\right],$$

where θ is the bifurcation angle. This expression originates from the geometrical analysis of particle trajectories in curved streamlines.

Simplified form in the ICRP model

ICRP Publication 66 uses an empirical formula fitted to experimental data to represent the contribution of inertial impaction:

$$\eta_{\text{imp}} = 1 - \exp(-\alpha \cdot d_a^2 \cdot Q),$$

where d_a is the aerodynamic diameter, Q is the inspiratory flow rate, and α is a region-specific empirical coefficient (e.g., for the tracheobronchial region $\alpha \approx 0.1$). This form originates from nonlinear regression of extensive human experimental data.

Mathematical derivation of gravitational sedimentation

Physical mechanism

Particles settle slowly onto airway walls under the influence of gravity. This mechanism dominates for **1–8 μm particles**, mainly occurring in the small airways and alveolar region (where airflow velocity is low and residence time is long). The size range of our engineered bacteria is **1–3 μm**, so **gravitational sedimentation is the core mechanism**.

Basic equation: Stokes settling velocity

In a quiescent fluid, the terminal settling velocity of a spherical particle at equilibrium between gravity and drag is given by Stokes' law.

Gravity:

$$F_g = m_p g = \frac{\pi}{6}\rho_p d_p^3 g.$$

Stokes drag (under laminar flow conditions):

$$F_d = 3\pi\mu d_p v_s.$$

Equating $F_g = F_d$ yields:

$$v_s = \frac{\rho_p d_p^2 g}{18\mu} \cdot C_c,$$

where ρ_p is the particle density, d_p is the particle diameter, g is the gravitational acceleration, μ is the dynamic viscosity of air, and C_c is a correction factor (approximately 1).

Derivation of settling probability in a horizontal airway

Consider a horizontal circular tube of length L and radius R (modeling a bronchiole or alveolar duct), with an average airflow velocity U, and the particle transit time $t = L/U$.

During time t, the particle settles a vertical distance $v_s t$. Assuming a uniform entry distribution, only those particles whose initial height (distance from the bottom) is less than $v_s t$ will deposit on the wall before the exit.

In a circular cross-section, the settling probability equals the ratio of the shaded area to the total cross-sectional area:

$$\eta_{\text{sed}} = \frac{2}{\pi R^2} \int_0^{v_s t} 2\sqrt{R^2 - (R-h)^2}\, dh.$$

Integration yields the exact expression:

$$\eta_{\text{sed}} = 1 - \frac{2}{\pi} \left[\sqrt{1 - \left(\frac{v_s t}{R}\right)^2} \left(1 - \frac{v_s t}{2R}\right) + \frac{1}{2} \arcsin\left(1 - \frac{v_s t}{R}\right) \right].$$

For small settling distances ($v_s t \ll R$), an approximation can be made and an exponential correction introduced to better match experimental data:

$$\eta_{\text{sed}} = 1 - \exp\left(-\frac{2}{\pi} \cdot \frac{v_s L}{UR}\right).$$

For the alveolar region, considering multiple breaths and mixing effects, it is further simplified to a form related to flow rate:

$$\eta_{\text{sed}} = 1 - \exp(-\beta \cdot d_a^2/Q),$$

where β is a region-specific empirical coefficient and Q is the inspiratory flow rate.

Mathematical derivation of diffusion deposition

Physical mechanism

Submicron particles (<0.5 μm) are influenced by Brownian motion and randomly collide with airway walls, leading to deposition.

Diffusion coefficient

Given by the Stokes–Einstein equation:

$$D_{\text{diff}} = \frac{k_B T C_c}{3\pi\mu d_p},$$

where k_B is Boltzmann's constant and T is the absolute temperature.

Expression for diffusion deposition efficiency

For laminar flow in a circular tube, the deposition efficiency can be expressed in series form:

$$\eta_{\text{diff}} = 1 - \sum_{i=1}^{\infty} a_i \exp(-b_i\Delta), \quad \Delta = \frac{\pi D_{\text{diff}} L}{4Q},$$

where a_i, b_i are series expansion coefficients.

3 Analysis of deposition mechanisms for engineered bacteria

The size range of our engineered bacteria is **1–3 μm**, which falls within the dominant regime of gravitational sedimentation. Therefore, in subsequent modeling, **gravitational sedimentation is the core mechanism**, while inertial impaction and diffusion deposition can be neglected unless the bacteria are encapsulated in nebulized droplets forming aggregates >5 μm.

Specifically, for 1–3 μm particles, the Stokes number is typically much less than 0.1 (for example, under typical upper airway conditions, d_p =

$3\,\mu$m, $U = 10\,$m/s, $D = 0.5\,$cm, $\rho_p = 1000\,$kg/m^3, yielding $Stk \approx 0.01$), so inertial impaction efficiency is extremely low. The diffusion mechanism is also weak for this size range, because the diffusion coefficient is inversely proportional to particle diameter; a 1 μm particle has a diffusion coefficient about two orders of magnitude smaller than that of a 0.1 μm particle.

4 Fitting form and application of the ICRP model

For engineering convenience, the ICRP model fits regional deposition fraction curves using **logistic functions (Sigmoid)** of the form:

$$y = \frac{A}{1 + \exp(-B(\ln d_a - C))}.$$

Because the deposition efficiency varies with the logarithm of particle diameter in an S-shaped manner (decrease at small sizes dominated by diffusion, increase at intermediate sizes due to sedimentation/impaction, and decrease again at large sizes due to upper airway interception), actual data are often represented by a superposition of two S-shaped curves. Thus, ICRP Publication 66 frequently uses the sum of two logistic terms:

$$DE_{\text{region}}(d_a) = \frac{A_1}{1 + \exp(-B_1(\ln d_a - C_1))} + \frac{A_2}{1 + \exp(-B_2(\ln d_a - C_2))}.$$

In addition, a **regional intake fraction factor** IF_{region} must be multiplied to account for the fact that not all inhaled particles reach that region due to upper airway interception. The parameters A, B, C etc. are determined through theoretical calculations and experimental data.

5 References

1. ICRP, 1994. *Human Respiratory Tract Model for Radiological Protection.* ICRP Publication 66.

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4. Hinds, W.C., 1999. *Aerosol Technology.*

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